



5.2 Optimisation with technology

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Maxima and minima of applied functions can be found using a GDC's graphical maximum/minimum finder, or by calculus (solve $f'(x)=0$ and classify with the second derivative test or sign change of $f'(x)$). A constraint (e.g. fixed perimeter or volume) is often used to write the quantity to be optimised as a function of a single variable before finding its extreme value, and any solution must be checked against the physically valid domain.

Worked example: follow the reasoning

A farmer has 200 m of fencing to enclose a rectangular field against an existing wall, so only 3 sides need fencing. Let x be the width perpendicular to the wall. Find the dimensions that maximize the enclosed area, and the maximum area.

Solution: With length L parallel to the wall, $2x+L=200$, so $L=200-2x$. Area

$A(x)=x(200-2x)=200x-2x^2$. $A'(x)=200-4x=0 \rightarrow x=50$, so $L=100$. Since $A''(x)=-4<0$, this is a maximum: $A=50 \times 100=5000 \text{ m}^2$.

Quick check · multiple choice

A GDC's graphical 'maximum' finder for a function $f(x)$ on an interval locates:

- A The y-intercept
- B The point where $f'(x)=0$ and f is a local maximum
- C The roots of $f(x)$
- D The inflection point

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A cylindrical can must have a volume of 500 cm^3 . Its total surface area (including top and bottom) is $S(r)=2\pi r^2+1000/r$, where r is the radius in cm. (a) Use technology or calculus to find the radius that minimizes the surface area, to 2 decimal places. (b) Find the minimum surface area, to the nearest cm^2 . (c) Find the height of the can at this radius, to 2 decimal places. [7]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $S'(r) = 4\pi r - 1000/r^2 = 0 \rightarrow r^3 = 1000/(4\pi) \approx 79.58 \rightarrow r \approx 4.30$ cm. (b) $S(4.30) = 2\pi(4.30)^2 + 1000/4.30 \approx 116.2 + 232.6 \approx 349$ cm². (c) $V = \pi r^2 h = 500 \rightarrow h = 500/(\pi \times 4.30^2) \approx 8.61$ cm.

Quick check answer

B. The maximum finder locates the local maximum, a stationary point where $f'(x) = 0$ with $f''(x) < 0$ (or an equivalent sign change).

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.